

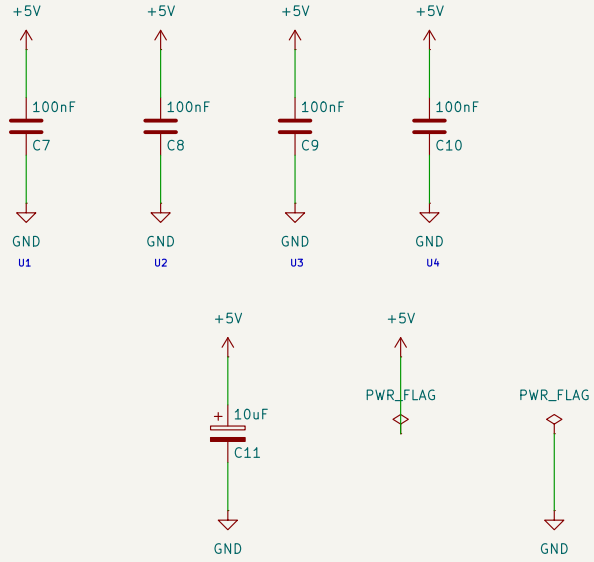
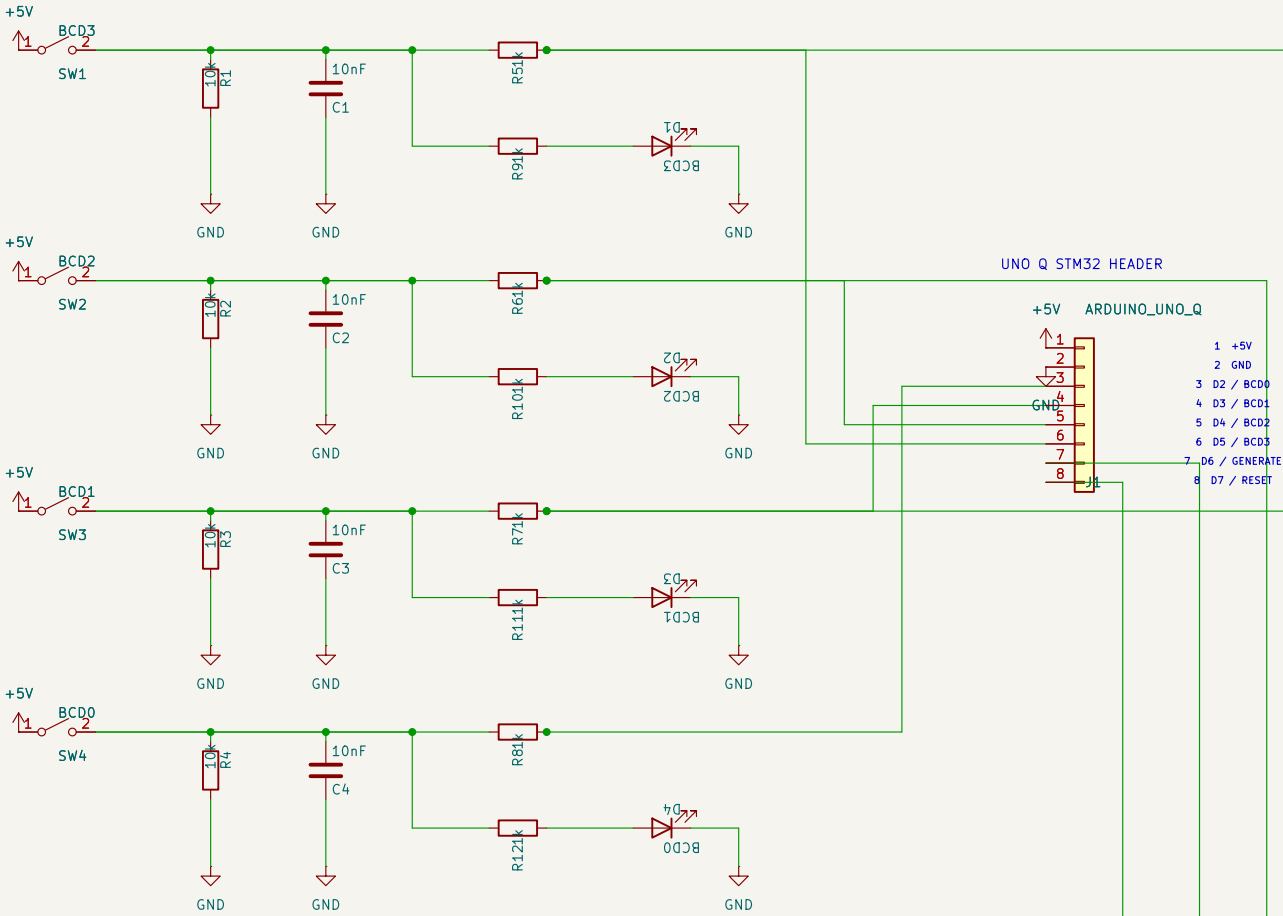
DIFFUINO – COMPLETE BCD INPUT AND CONTROL CIRCUIT

Hardware validity: VALID = NOT [BCD3 AND (BCD2 OR BCD1)]

1. BCD SELECTOR, INPUT FILTERS, AND VISIBLE BIT INDICATORS

5. LOCAL SUPPLY DECOUPLING

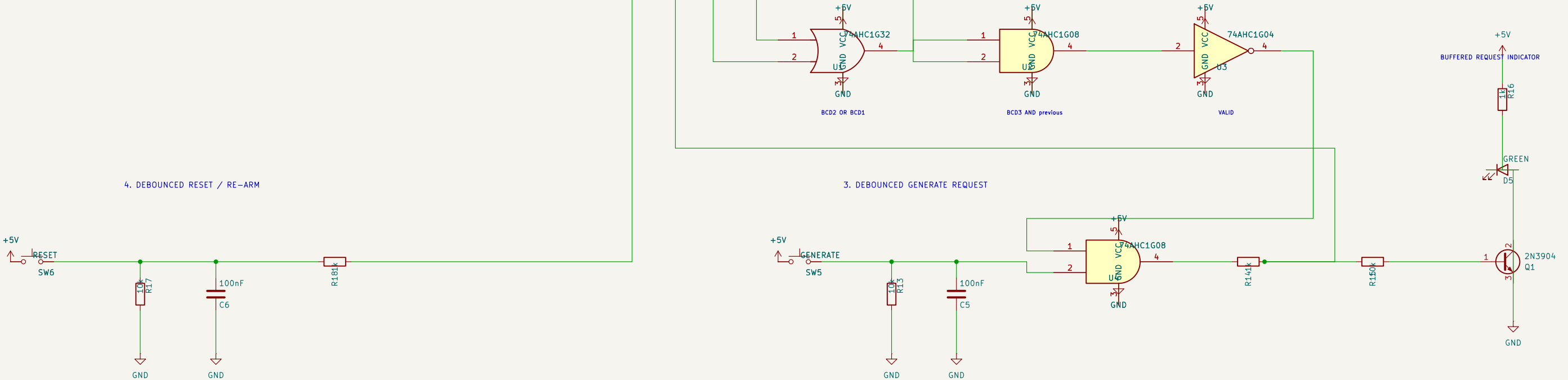
ON = +5 V OFF = 10 kOhm pull-down



2. HARDWARE DECIMAL-VALIDITY INTERLOCK

3. DEBOUNCED GENERATE REQUEST

4. DEBOUNCED RESET / RE-ARM



D2–D7 only; do not use 1.8 V MPU GPIO Arduino UNO Q / STM32 digital interface MonoDiffuse		
Sheet: / File: diffuino_selector.kicad_sch		
Title: Diffuino BCD selector, validity interlock, and reset		
Size: A3	Date:	Rev: 2.0
KiCad E.D.A. 10.0.5		Id: 1/1

Firmware: 40 ms debounce; BCD is latched for inference; D6 is ignored while busy; D7 restores the ready smiley.

D2–D7 are the UNO Q's 5 V-tolerant STM32 header inputs. Never connect this circuit to the 1.8 V MPU GPIO.